

Attention Propagates, the FFN Overwrites: A Relational Correction and Generalization of the Peak-Relative Flip

Working note, version 2 — supersedes the interpretation (not the measurements) of “The
FFN Flip is Peak-Relative” (v1)

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July 4, 2026

Abstract

Version 1 of this note reported, with pre-registered confirmation on three models across two families, that the feed-forward network writes pro-truth into the block where the truth axis peaks and flips to anti-truth at exactly peak+1. An axis-provenance control run after publication — fitting the measuring axis at several different blocks — shows that this reading was the special case of a more general law, contaminated by a measurable circularity. The corrected, fully general statement, confirmed by pre-registered prediction on *four* models across two families (Qwen2.5-1.5B/3B, Llama-3.2-1B/3B): **the FFN’s contribution at block b is anti-aligned with every truth frame that does not contain it** (the pre-FFN frame of its own block, and all earlier frames) **and pro-aligned only with frames that contain it; attention shows the mirror image**, pro-aligned with existing frames it did not write. At the peak, under the clean pre-FFN ruler, the two components are near mirror images (e.g. attention +1.63 vs. FFN −1.43 on Llama-3.2-3B). We name the asymmetry *propagate vs. overwrite*: attention transports the existing truth frame; the FFN opposes it and deposits the material of the next one — the axis is not an object the FFN erodes but a moving summary the FFN’s writes constitute. The correction separates two phenomena that v1 conflated: frame-relative rotation (what the “flip” actually was) and frame-free signal decay (the genuine erosion, which survives untouched: zero-ablating the FFN still raises *refit* truth AUC, and freezing its value-stream variation still recovers it). The knowledge-dependent dimensionality law gains a fourth point (behavioral KNOWN rate 42% → measured gap 0.112 on Llama-3.2-3B, inside the registered window), and the “ignition” of the truth axis is decided by pre-registered prediction to be a family trait, not a scale effect. All v1 numbers remain valid and reproducible; what changed is what one of them meant. This is a versioned correction, not a retraction, and the note documents the control that forced it, because the failure was bought with a pre-registered discrimination and is therefore worth more than the claim it replaced.

1 Scope, honesty, and what changed from v1

Everything in v1 that was a *measurement* stands: every number reproduces. What fell is one *interpretation*: “the flip is anchored to peak+1.” Section 4 narrates how it fell, because the sequence — claim, self-raised objection against the instrument, pre-registered control, exposure, generalization — is the useful content for a reader, not an embarrassment to be footnoted. For the impatient, the status ledger:

- **Reinterpreted (this note):** the peak-anchored flip → the relational law of Sec. 5, plus a measured circularity (Sec. 6).

- **Untouched from v1:** the causal erosion (frame-free; Sec. 7); the value-stream attribution (causal freezes are frame-free; the observational split table requires frame-aware rereading); the knowledge-dependent dimensionality law (now four points); the pre-registered negative on the static OV/QK crossover; the late-layer decoupling.
- **New in v2:** the axis-provenance method and its synthetic validation; the relational law on four models; the propagate/overwrite asymmetry; the fourth dimensionality point; ignition as a family trait.

Evidential levels as in v1: Level A = cross-model with predictions written before the runs (everything in Secs. 5–8); Level B = single-model causal (carried from v1, Sec. 7); Level C = declared leads (Sec. 9).

2 Setup

Four base models, two families:

model	blocks	d_{model}	truth peak p	rel. depth
Qwen2.5-1.5B	28	1536	15	0.54
Qwen2.5-3B	36	2048	16	0.44
Llama-3.2-1B	16	2048	7	0.44
Llama-3.2-3B	28	3072	9	0.32

Dataset, axis construction, held-out protocol, precision policy (float32 for all decomposition runs, gated by the additive identity check; median relative errors $0.65\text{--}1.17 \times 10^{-7}$ across the runs of this note), and conventions (“block b ” output = `hidden_states[b+1]`) are exactly those of v1 and the companion paper. The spread of the four peaks in relative depth (0.32–0.54) independently reinforces the v1 negative on the static circuit crossover.

3 Method: the axis-provenance control

V1’s central instrument was the *fixed-axis contribution gap*: fit the truth axis v_1 once at the peak block (training folds only), project every component’s contribution at every layer onto it, and report the held-out class gap d' . The objection — raised by the author, against the author’s instrument — is that a flip anchored to “peak+1” could be a property of *where the ruler is fit* rather than of the FFN. The control: fit v_1 at several blocks $b \in \{p-4, p-2, p, p+2\}$ and recompute the full gap profile under each ruler. Two rulers per fit block, from the same cached tensors:

post : v_1 fit on the block output $h_{b+1} = h_b + a_b + f_b$ (contains f_b),
pre : v_1 fit on the pre-FFN state $h_b + a_b$ (f_b excluded).

The pre ruler removes the direct circularity of measuring f_b against an axis fit on a state that contains f_b . The discrimination criterion, fixed before the runs and validated on synthetic ground truth (a planted global axis with a planted writer flip: the measured transition stays pinned under every ruler; a planted circular writer collapses under the pre ruler with ratio 0.04 while planted genuine alignment survives at 0.99): genuine alignment predicts the pre-ruler diagonal stays clearly positive; circularity predicts collapse. A methodological note learned the hard way: two successive automatic “transition-rule” verdicts in the tooling misread these matrices during the analysis; the released tool prints the matrices and diagonal summaries with *no* auto-verdict, and all readings below are of the full matrices.

4 From peak-anchored to relational: how the v1 reading failed, and what the failure bought

Act 1 (v1). The FFN gap, measured under the peak-fit ruler, is pro-truth at p and anti-truth at $p+1$, replicated with pre-registered predictions on three models across two families. The measurement was correct; the anchor was not yet tested.

Act 2 (the control bites its owner). Under rulers fit at $p-4, p-2, p, p+2$, the transition does not stay at $p+1$: it *tracks the ruler*. On Qwen2.5-1.5B the diagonal reads pro at b / anti at $b+1$ for $b = 13, 15, 17$ alike (+0.61 / - 0.79, +0.57 / - 0.82, +0.52 / - 0.73); on Qwen2.5-3B for $b = 16, 18$ (+0.69 / - 1.23, +0.73 / - 0.79); later on both Llamas (Table 1). The peak-anchored flip was the $b=p$ slice of a relational pattern present at every block where an axis exists.

Act 3 (the pre ruler exposes the circularity). Under the pre-FFN ruler the “pro at own block” term does not merely collapse — it *inverts sign* (Table 2): measured against the frame that exists at the moment it writes, the FFN’s contribution is strongly anti-aligned. The pro term of the post ruler was, in substantial part, the ruler measuring its own ingredient (f_b is inside h_{b+1} , and drags the axis fit there toward itself); the residual above-diagonal positives under the pre ruler (e.g. f_{13} at +0.48 / +0.60 on pre-frames of blocks 15/17 in Qwen-1.5B) are the same circularity one step forward, since those later frames contain f_{13} . The fully clean statement survives every containment check and is the law of the next section. What the failure bought: a general law instead of a special case, a measured mechanism (Sec. 6) instead of a coordinate, and a live demonstration that the shields bind their owner.

5 Level A: the relational law

Table 1: POST-ruler diagonals on four models: per fit block b , the FFN’s held-out d' at b and at $b+1$. Pro at b , anti at $b+1$, for every b where an axis exists — not only at the peak.

model (peak)	ruler b	$d'_{\text{ffn}}(b)$			$d'_{\text{ffn}}(b+1)$
Qwen2.5-1.5B (15)	13 / 15 / 17	+0.61 /	+0.57 /	+0.52	-0.79 / -0.82 / -0.73
Qwen2.5-3B (16)	14 / 16 / 18	+0.06 /	+0.69 /	+0.73	-0.09 / -1.23 / -0.79
Llama-3.2-1B (7)	5 / 7 / 9	+0.13 /	+0.54 /	+0.19	-0.07 / -0.95 / -0.54
Llama-3.2-3B (9)	7 / 9 / 11	+0.37 /	+0.49 /	+0.57	-0.42 / -1.25 / -1.39

Where the axis is weak or absent, the pattern correctly vanishes: rulers fit before the Qwen-3B ignition (fit-block cosines to the peak axis 0.049 and 0.079) and before the Llama-1B mid-knowledge peak return noise — the internal consistency expected if the law is about frames that exist.

6 Circularity exposed, and the propagate-overwrite asymmetry

Table 2: PRE-ruler diagonals: the same cells with f_b excluded from the ruler’s data. The FFN inverts sign (circularity exposed; it writes *against* the frame of its moment). Attention, measured identically, is the mirror image.

model	ruler b	$d'_{\text{ffn}}(b)$			$d'_{\text{attn}}(b)$
Qwen2.5-1.5B	13 / 15 / 17	-0.63 /	-0.41 /	-0.78	+1.56 / +1.31 / +1.04
Qwen2.5-3B	14 / 16 / 18	-0.14 /	-1.16 /	-0.46	+0.26 / +1.38 / +1.40
Llama-3.2-1B	7 / 9	-1.17 /	-0.82		+1.26 / +0.82
Llama-3.2-3B	7 / 9 / 11	-0.81 /	-1.43 /	-1.41	+0.99 / +1.63 / +1.75

The decisive zone is the sub-diagonal of the pre matrices — contributions at layer L measured against frames of blocks strictly before L , which contain neither f_L nor a_L and are therefore clean of every containment effect. There (Fig. 1) the FFN is negative nearly everywhere an axis exists, and attention is positive nearly everywhere an axis exists, on all four models. At the peak, under the same clean frame, the two components are near mirror images: $+1.31/-0.41$ (Qwen-1.5B), $+1.38/-1.16$ (Qwen-3B), $+1.26/-1.17$ (Llama-1B), $+1.63/-1.43$ (Llama-3B). Pre-registered predictions for the two Llamas (written before those runs): POST pro/anti diagonal; PRE sign inversion for the FFN; sub-diagonal positive for attention and negative for the FFN. All confirmed.

We name the asymmetry **propagate vs. overwrite**. Attention, which reads the residual stream, transports the truth frames already present — it is pro-aligned with directions it did not write. The FFN, which generates content from its weights, opposes the frame of its moment and deposits the material of the next: its writes are what later frames are made of. The correct ontology dissolves v1’s implicit picture of an axis being eroded: *the axis is not an object in the model; it is a moving statistical summary of where the class difference lives, and the FFN’s class-signed writes constitute it*. This is the dynamical face of the key–value write-head physiology attributed to MLPs by the memory-editing literature [6], and it supplies a candidate microscopic writer for the macroscopic finding of [7] that factual selection separates by rotation, not rescaling, with active suppression of the correct continuation: the component that opposes the current frame and writes the next one is, per these measurements, the FFN.

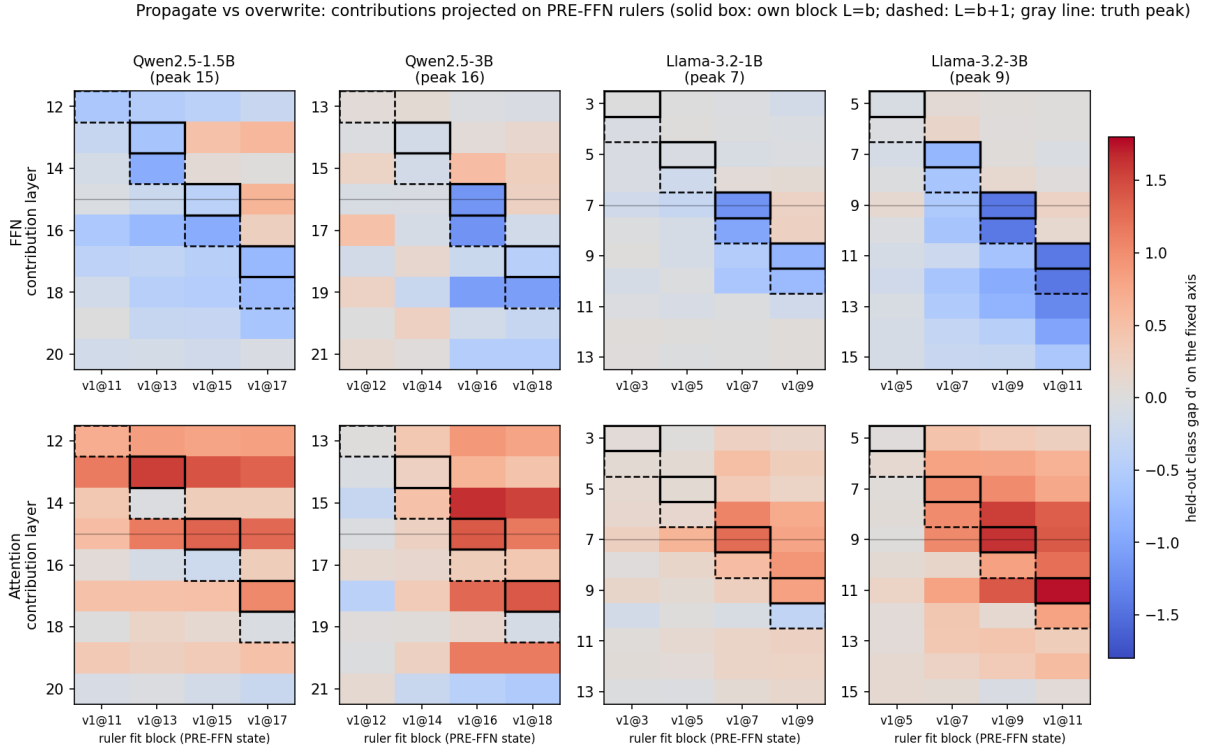


Figure 1: Propagate vs. overwrite on four models. Held-out class gap d' of each component’s contribution (rows: FFN above, attention below) projected on PRE-FFN rulers fit at four blocks (columns within each panel). Solid box: own block $L=b$; dashed: $L=b+1$; gray line: truth peak. The FFN row is cold wherever an axis exists; the attention row is warm.

7 What the correction does not touch: erosion is real and frame-free

Rotation alone cannot explain two v1 measurements that use no frozen ruler. First, the *refit* truth AUC (axis re-fit at every layer) decays after the peak (e.g. $0.79 \rightarrow 0.54$ on Qwen-1.5B): if the FFN only rotated the frame while preserving class signal, refit readability would not fall. Second, the causal results: zero-ablating the FFN over the post-peak band *raises* refit AUC (v1: $+0.103$ on Qwen-1.5B, $+0.053$ on Qwen-3B), attention-off does not, and freezing only the value-stream variation recovers most of the effect ($+0.089$ vs. $+0.040$ for the gate). The v2 picture therefore separates two phenomena that v1 conflated under “erosion”: (i) frame-relative rotation — what the peak-anchored flip actually was — causally driven by the FFN (intact entry→readout axis cosines $0.627/0.667$ vs. $0.876/0.890$ with the FFN off); and (ii) frame-free decay of class signal in the late band — the genuine erosion, which stands on its own causal evidence and is carried by the value stream.

8 The dimensionality law’s fourth point, and ignition as a family trait

model	KNOWN rate	axis AUC	probe AUC	gap
Qwen2.5-3B	(high; see v1)	0.832	0.881	0.049
Llama-3.2-3B	42% (measured first)	0.785	0.897	0.112
Llama-3.2-1B	42% (measured first)	0.690	0.831	0.141
Qwen2.5-1.5B	(low-mid; companion paper)	0.638–0.795	0.841	≈ 0.20

For Llama-3.2-3B the expectation (mid knowledge $\rightarrow$ intermediate gap) was registered from the behavioral check before the probe was run; the measured 0.112 fell inside the window. Four models, two families: the axis-vs-probe gap tracks measured knowledge every time.

Ignition. V1 left open whether the Qwen-3B “switch-on” of the axis (per-layer AUC $0.526 \rightarrow 0.790$ in one layer; fit-block cosines $0.049/0.079$ before the peak) is a scale effect or a family trait. The pre-registered discrimination: scale predicts a step in Llama-3.2-3B; family predicts a ramp. Measured: a ramp ($0.526 \rightarrow 0.657 \rightarrow 0.664 \rightarrow 0.749 \rightarrow 0.785$ across levels 6–10). Ignition is, on present evidence, a trait of the Qwen recipe at the 3B scale, not of scale itself.

9 Leads, declared as leads

(i) On both mid-knowledge Llamas, phase/second-dimension scores match or beat the single axis off-peak (Llama-1B: 2D $+0.092$ held-out at the peak; Llama-3B: PHASE $+0.08$ – 0.10 in the decay zone) — the mid-regime second dimension of v1’s leads, still awaiting a dedicated null. (ii) The per-head version of the asymmetry: whether heads externally labeled as payload-delivery (“late induction”) show FFN-like overwrite signatures while the majority propagates — the natural joint experiment with the HeadGenome taxonomy [8]. (iii) The observational gate/value split table of v1 should be re-read against pre-FFN rulers; the causal attribution stands regardless.

10 Limitations

Four models, still two families; both additions are Llamas, so family breadth did not grow in v2. The pre ruler removes the direct containment circularity but not indirect paths (accumulated earlier writes shape the pre-frame). The relational law is measured on last-token states over one dataset (CounterFact, pinned revision). The propagate/overwrite reading is a description of measured alignments, not yet a circuit-level mechanism; *why* attention inherits frames and the

FFN opposes them is the open question this note sharpens rather than answers. Sub-diagonal cells before an axis exists are uninformative by construction and were predicted to be noise (they are). No permutation nulls were run on the provenance matrices themselves; the diagonal magnitudes ($|d'|$ up to 1.6) sit far above the null scales measured on identical statistics in v1, but the formal check is future work.

11 Reproducibility

One new tool, `axis_provenance.py` (imports `truth_probe.py`, `anatomy.py`, `flip_consolidate.py`; prints matrices and diagonal summaries, no auto-verdict; synthetic ground-truth and collapse-ratio validations included in the repository’s test files). All v1 tools unchanged.

```
# per model: FFN and attention, post+pre rulers, one extraction each
python axis_provenance.py --peak 15 --scan-start 12 --scan-end 20
python axis_provenance.py --peak 15 --scan-start 12 --scan-end 20 --component attn
python axis_provenance.py --model Qwen/Qwen2.5-3B --peak 16 \
    --scan-start 13 --scan-end 21          # (+ --component attn)
python axis_provenance.py --model meta-llama/Llama-3.2-1B --peak 7 \
    --scan-start 3 --scan-end 13          # (+ --component attn)
python axis_provenance.py --model meta-llama/Llama-3.2-3B --peak 9 \
    --scan-start 5 --scan-end 15          # (+ --component attn)

# fourth dimensionality point (behavior BEFORE geometry)
python behav_check.py --model meta-llama/Llama-3.2-3B
python truth_probe.py signal --dataset counterfact --max-pairs 250 \
    --model meta-llama/Llama-3.2-3B --baseline --perm 100 \
    --rev-counterfact c945b082ca08d0a8f3ba227fb78404a09614c36e
```

Upload `provenance_asymmetry.png` alongside this `.tex`. License: CC BY 4.0. V1 remains available as the previous version of this record; its figures are the $b=p$ slice of the matrices reported here.

References

- [1] F.K. Vicidomini. *How Much of Truth Fits on a Single Axis?* Working paper, June 2026. Zenodo DOI 10.5281/zenodo.20938285.
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